TUO Open Problems: Numerical Analysis and Next Steps

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Abstract

The Theory of Universal Origins (TUO) proves fourteen results from two axioms, but explicitly sets aside six open problems. This note works through each problem numerically, identifying (a) what can be computed from known physics today, (b) what quantity or calculation would close each problem, and (c) a new result: the thermalisation time $\tau_{\rm th}$ with realistic FRW cooling is $152\,t_{\rm Pl}$, revising the junction range from $[17, 94]\,t_{\rm Pl}$ to $[17, 152]\,t_{\rm Pl}$.

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1 Numerical Values Used Throughout

All calculations use:

$$g_* = 106.75, \quad T_{\rm TUO} = (15/\pi^2)^{1/4}\, T_{\rm Pl} = 1.1103\, T_{\rm Pl}, \quad \tau_{gg} = 1.489\, t_{\rm Pl}, \quad \alpha_s\!\left(\tfrac{E_{\rm Pl}}{2}\right) = 0.01055.$$

The six open problems from the paper are OP1 (baryon asymmetry), OP2 (CMB power spectrum), OP3 (dark energy), OP4 (precise junction time), OP5 (SM content from axioms), and OP6 (SM gauge group).

2 OP1 — Baryon Asymmetry η

Open Problem 1

TUO shows that $B - L = 0$ permits matter-only configurations. It does not derive $\eta = n_B/n_\gamma = 6.12 \times 10^{-10}$.

2.1 What TUO gives

At emergence, the maximum fluctuation has

$$B = L = 6 \quad (3 \text{ generations} \times 2 \text{ per gen, no antiparticles}).$$

The photon number density in the thermal bath at $T_{\rm TUO}$ is

$$n_\gamma = \frac{2\zeta(3)}{\pi^2}\left(\frac{k_{\rm B} T_{\rm TUO}}{\hbar c}\right)^3,$$

giving $N_\gamma^{\rm cell} = n_\gamma\, \ell_{\rm Pl}^3 = 0.3334$ photons per Planck cell. A naive ratio gives

$$\eta_{\rm naive} = \frac{B}{N_\gamma^{\rm cell}} = \frac{6}{0.333} \approx 18,$$

which is $\sim 3 \times 10^{10}$ times the observed value. This immediately shows that most of the initial baryon number is washed out between emergence and nucleosynthesis.

2.2 The sphaleron problem

Electroweak sphalerons are in equilibrium for $T > T_{\rm EW} \approx 100\,\text{GeV}$. Sphalerons violate $B + L$ but conserve $B - L$. Since $(B - L)_{\rm initial} = 0$ exactly (TUO axiom), sphalerons drive

$$B + L \;\to\; 0 \quad (\text{since } B - L = 0 \Rightarrow B = L),$$

leaving $B_{\text{final}} = 0$ and therefore $\eta = 0$ at sphaleron equilibrium.

Finding

TUO with exact $B - L = 0$ predicts $\eta = 0$ at leading order. This is consistent: a matter-only universe requires a *sub-leading* source of $B - L \neq 0$ that the leading-order axiom does not capture.

2.3 What is needed to close OP1

The observed $\eta = 6.12 \times 10^{-10}$ requires

$$\delta(B - L) \equiv (B - L)_{\text{sub-leading}} = \eta\, N_\gamma^{\text{cell}} = 2.04 \times 10^{-10} \text{ per cell},$$

which is $\delta(B - L)/B_{\text{initial}} = 3.4 \times 10^{-11}$. Two mechanisms can generate this:

1. **Leptogenesis.** If right-handed neutrinos with Majorana masses $M_R \lesssim T_{\text{TUO}}$ exist, their CP-violating out-of-equilibrium decays at $T < T_{\text{TUO}}$ generate δL, which sphalerons convert to δB. The required CP asymmetry is

$$\varepsilon_{\text{CP}} \sim \frac{\eta\, N_\gamma^{\text{cell}}}{B_{\text{initial}}} \sim 3.4 \times 10^{-11}.$$

   This is achievable for $M_R \sim 10^{12}$ GeV and $\mathcal{O}(1)$ Yukawa couplings (Davidson-Ibarra lower bound).

2. **Sub-leading path-integral term.** The TUO constrained path integral enforces $B - L = 0$ at the saddle point. A next-to-leading saddle could carry $\delta(B - L)$ suppressed by the instanton action $S_{\text{inst}} \sim 2\pi/\alpha_s \approx 600$:

$$\delta(B - L) \sim e^{-S_{\text{inst}}} \approx e^{-600} \ll \eta.$$

   This is far too small. Mechanism 1 (leptogenesis) is therefore the more natural path.

What closes OP1

Compute the leptogenesis yield $\eta(\varepsilon_{\text{CP}}, M_R, T_{\text{TUO}})$ using the standard Boltzmann equations for N_R decay, sourced by the TUO initial thermal bath at $T_{\text{TUO}} = 1.1103\, T_{\text{Pl}}$. The target is $\varepsilon_{\text{CP}} \approx 3.4 \times 10^{-11}$, which constrains M_R and the right-handed neutrino sector.

3 OP2 — CMB Power Spectrum

Open Problem 2

The saddle-point argument suggests simultaneous emergence and $\Delta T/T = 0$ at leading order. Within-cell fluctuations give $\delta E/E \sim 1/\sqrt{g_*}$. The observed $\Delta T/T \sim 10^{-5}$ requires $\sim 10^3$ suppression.

3.1 Cell-level fluctuation budget

Each Planck cell contains $g_* = 106.75$ independent modes. By central limit theorem, energy fluctuations are

$$\frac{\delta E_{\text{cell}}}{E_{\text{cell}}} = \frac{1}{\sqrt{g_*}} = 0.0968 \quad (9.7\%).$$

Using the Stefan-Boltzmann relation $E \propto T^4$:

$$\frac{\delta T}{T} = \frac{1}{4}\frac{\delta E}{E} = \frac{1}{4\sqrt{g_*}} = 0.0242 \quad (2.4\%).$$

Compared to the observed $\Delta T/T \sim 10^{-5}$, the required suppression factor is

$$\mathcal{S} = \frac{\delta T/T|_{\text{cell}}}{\Delta T/T|_{\text{obs}}} \approx \frac{0.024}{10^{-5}} = 2.4 \times 10^3.$$

3.2 Sources of suppression

Three GR mechanisms reduce cell-scale fluctuations to CMB-scale:

1. **Jeans suppression.** The Jeans length at the Planck epoch:

$$\lambda_J = c_s\sqrt{\frac{\pi}{G\rho/c^2}}, \qquad c_s = c/\sqrt{3},$$

   gives $\lambda_J = 0.14\,\ell_{\text{Pl}}$. All modes with $\lambda > \lambda_J$ are sound-wave-suppressed during radiation domination. This does not by itself give 10^{-5}, but sets the tilt of the initial power spectrum.

2. **Silk damping.** Photon diffusion on scales $\lambda \ll \lambda_{\text{mfp}}$ damps fluctuations exponentially. Silk scale at decoupling: $\lambda_{\text{Silk}} \approx 8\,\text{Mpc}$.

3. **Super-Hubble freezing.** Modes larger than the Hubble radius at Planck epoch freeze as they exit the sound horizon. For a cell of size ℓ_{Pl} at $t = t_{\text{Pl}}$, the ratio $\delta T/T|_{\text{frozen}} \sim (\ell_{\text{Pl}}/H^{-1})^2 = (t_{\text{Pl}}\,H)^2 = 1/4$ picks up an additional $(1/4)^n$ suppression per horizon crossing.

Finding

The observed $\Delta T/T \sim 10^{-5}$ needs suppression from $\delta T/T|_{\text{cell}} = 2.4\%$ by a factor of ~ 2400. This is within reach of a GR propagation calculation: $\log_2(2400) \approx 11$ e-folds of suppression through horizon crossing from ℓ_{Pl} to the last-scattering surface.

What closes OP2

Compute the TUO two-point correlator

$$\mathcal{P}_\zeta(k) = \frac{k^3}{2\pi^2} \int d^3r\, e^{-i\mathbf{k}\cdot\mathbf{r}} \left\langle \hat{T}_{00}(\mathbf{x})\, \hat{T}_{00}(\mathbf{x}+\mathbf{r}) \right\rangle_{\text{ZS}}$$

at the Planck-cell scale in the zero-sum constrained vacuum, then propagate to the last-scattering surface using standard Boltzmann transfer functions. The spectral tilt n_s is the primary observable output.

4 OP3 — Residual Dark Energy Λ

Open Problem 3

TUO predicts $\Lambda_{\text{eff}} = 0$ at tree level (zero-sum axiom). The observed $\rho_\Lambda \approx 5.4 \times 10^{-10}\,\text{J/m}^3$ is not derived.

4.1 The cosmological constant in TUO language

The energy densities involved:

Quantity	Value (J/m^3)	Ratio to ρ_{Pl}
Planck density $\rho_{\text{Pl}} = E_{\text{Pl}}/\ell_{\text{Pl}}^3$	4.63×10^{113}	1
1-loop QFT estimate $\sim (g_*/16\pi^2)\ln(E_{\text{Pl}}/E_{\text{EW}})\rho_{\text{Pl}}$	$\sim 27\,\rho_{\text{Pl}}$	~ 27
Observed ρ_Λ	5.4×10^{-10}	1.16×10^{-123}

4.2 TUO tree-level prediction

The zero-sum axiom gives $E_{\text{matter}} + E_{\text{grav}} = 0$ exactly. This means the zero-point energy of the vacuum is constrained to vanish at the leading saddle-point of the constrained path integral. Symbolically: $\langle\hat{\rho}_0|\hat{H}|\hat{\rho}_0\rangle = 0$ by Axiom II.

4.3 The gap

Standard quantum field theory contributes loop corrections

$$\rho_{\text{vac}}^{1-\text{loop}} \sim \frac{g_*}{16\pi^2} \ln\left(\frac{E_{\text{Pl}}}{E_{\text{EW}}}\right) \rho_{\text{Pl}} \approx 26.6\, \rho_{\text{Pl}}.$$

The zero-sum axiom must cancel this to $10^{-123}\rho_{\text{Pl}}$. Two scenarios:

1. **Exact cancellation + small residual.** The constrained PI enforces exact cancellation of all loop contributions by the gravitational field sector. The residual ρ_Λ arises from a *topological* term — analogous to the QCD θ-vacuum — that is exponentially suppressed but nonzero:

$$\rho_\Lambda \sim \rho_{\text{Pl}}\, e^{-S_{\text{top}}}, \quad S_{\text{top}} = \ln\big(\rho_{\text{Pl}}/\rho_\Lambda\big) \approx 283.$$

   A topological action of $\sim 283\,\hbar$ is not implausible for a Planck-scale gravitational instanton.

2. **Quintessence in TUO.** A slowly-rolling field ϕ with zero-sum charge $Q_\phi = 0$ is admitted by the axiom. Its potential energy contributes $\rho_\phi(t) \sim \rho_\Lambda(t)$ with appropriate dynamics.

What closes OP3

This is the cosmological constant problem in TUO language. The precise target is: find the next-to-leading saddle of the constrained path integral (Eq. 8, main paper) and compute its contribution to ρ_{vac}. The required suppression, e^{-283} relative to ρ_{Pl}, must emerge from the structure of the zero-sum constraint. This is the hardest open problem in TUO; it is also the hardest open problem in all of theoretical physics.

5 OP4 — Precise Junction Time

Open Problem 4

The junction range $t_{\text{junc}} \in [17,\, 94]\, t_{\text{Pl}}$ is established. The precise value requires a Boltzmann-equation treatment of the QGP with realistic cooling.

5.1 The two bounds established in the paper

Lower bound $t_{\text{class}} = \sqrt{300}\, t_{\text{Pl}} \approx 17.3\, t_{\text{Pl}}$**.** This is when $\delta V/V = 3(\ell_{\text{Pl}}/ct)^2 < 1\%$, i.e., the quantum correction to the expansion volume becomes negligible and a classical GR description is valid.

Upper bound (paper) $\tau_{\rm th} \approx 85$–$94\, t_{\rm Pl}$**.** The isothermal mean free path: $\lambda_{\rm mfp} = \hbar c/(\alpha_s k_{\rm B} T_{\rm TUO})$. This assumes $T = T_{\rm TUO}$ is held constant throughout — an overestimate of T at late times.

5.2 New result: thermalisation with FRW cooling

After the classical transition at $t_{\rm class}$, the universe enters radiation-dominated FRW with

$$T(t) = T_{\rm TUO}\sqrt{\frac{t_{\rm class}}{t}}, \quad t > t_{\rm class}.$$

The temperature drops, increasing the mean free path: $\lambda_{\rm mfp}(t) = \lambda_0\sqrt{t/t_{\rm class}}$ where $\lambda_0 = \hbar c/(\alpha_s k_{\rm B} T_{\rm TUO})$.

Thermalisation occurs when the optical depth $\int_0^{\tau_{\rm th}} c\, dt/\lambda_{\rm mfp}(t) = 1$. Splitting the integral into the isothermal phase $[0, t_{\rm class}]$ and the cooling phase $[t_{\rm class}, \tau_{\rm th}]$:

Phase 1 ($T = T_{\rm TUO}$, constant):

$$I_1 = \frac{c\, t_{\rm class}}{\lambda_0} = 0.2029.$$

Phase 2 ($T \propto t^{-1/2}$, cooling):

$$I_2 = \int_{t_{\rm class}}^{\tau_{\rm th}} \frac{c}{\lambda_0\sqrt{t/t_{\rm class}}}\, dt = \frac{2c\sqrt{t_{\rm class}}}{\lambda_0}\left(\sqrt{\tau_{\rm th}} - \sqrt{t_{\rm class}}\right).$$

Setting $I_1 + I_2 = 1$ and solving for $\tau_{\rm th}$:

$$\sqrt{\tau_{\rm th}} = \sqrt{t_{\rm class}} + \frac{(1 - I_1)\,\lambda_0}{2c\sqrt{t_{\rm class}}},$$

giving

$$\boxed{\tau_{\rm th} = 152.2\, t_{\rm Pl}.} \tag{1}$$

New result (OP4)

With realistic FRW cooling ($T \propto t^{-1/2}$ after $t_{\rm class}$), the thermalisation time is $\tau_{\rm th} = 152.2\, t_{\rm Pl}$, compared to the isothermal upper bound of 85–94 $t_{\rm Pl}$.
The corrected junction range is:

$$t_{\rm junc} \in [17.3,\ 152.2]\, t_{\rm Pl}.$$

The paper's $[17, 94]\, t_{\rm Pl}$ range is conservative (isothermal assumption). The realistic upper bound is $\approx 1.6\times$ larger.

5.3 Physical interpretation

The cooling extends the thermalisation time because particles slow down (longer mean free path) as the universe expands and cools. The QGP spends most of its time in the free-streaming regime even after the classical transition. Full thermalisation at $\sim 152\, t_{\rm Pl}$ means:

- Kinetic equilibrium (equal temperature distribution): at $\tau_{\rm th}$.
- Chemical equilibrium (stable particle ratios): somewhat later, $\tau_{\rm chem} \sim \tau_{\rm th} \times (1/\alpha_s^2) \sim \mathcal{O}(10^4\, t_{\rm Pl})$.

What closes OP4 fully

Replace the leading-order $\lambda_{\rm mfp} = \hbar c/(\alpha_s k_{\rm B} T)$ with the 2-loop QCD transport coefficient $\eta_{\rm shear}/s \approx 1/(4\pi)$ (KSS bound), and include inelastic QCD processes ($gg \leftrightarrow ggg$). The 2-loop result modifies $\tau_{\rm th}$ at the $\sim 20\%$ level, tightening the range to $t_{\rm junc} \approx [17, 120\text{–}180]\, t_{\rm Pl}$.

6 OP5 — SM Particle Content from Axioms

Open Problem 5

TUO uses $g_* = 106.75$ from experiment. Can the SM field content and $N_{\rm gen} = 3$ be derived from the constrained path integral?

6.1 What TUO constrains

The zero-sum filter (Axioms I+II) admits any field content satisfying:

(i) Anomaly cancellation: $Q = 0$, $B - L = 0$ per generation.

(ii) No antiparticles: a matter-only stable configuration exists.

Table 7.1 of the paper shows that configurations with $N_{\rm gen} = 1, 2, 3, 4, 5$ all satisfy these. TUO does *not* prefer $N_{\rm gen} = 3$ from these conditions alone.

6.2 Single-cell action for each configuration

The Euclidean action for a single-cell emergence is $S = E_{\rm cell}\, t_{\rm Pl}/\hbar = g_*/2$:

N_{gen}	g_*	$S/\hbar$	e^{-S}
1	58.0	29.0	2.5×10^{-13}
2	88.0	44.0	7.8×10^{-20}
3	**118.0**	**59.0**	$\mathbf{2.4 \times 10^{-26}}$
4	148.0	74.0	7.3×10^{-33}
5	178.0	89.0	2.2×10^{-39}

The single-cell action is minimised at $N_{\text{gen}} = 1$, so the unconstrained measure e^{-S} alone would select $N_{\text{gen}} = 1$.

6.3 The global constraint changes everything

For simultaneous emergence of all $N = 8.8 \times 10^{92}$ cells (the dominant saddle, Sec. 11 of main paper), the total action is $S_{\text{total}} = N \times S_{\text{single}} = Ng_*/2$. Selecting $N_{\text{gen}} = 1$ (minimum single-cell action) gives the largest e^{-NS} in the simultaneous sector.

However, the zero-sum global constraint couples cells: the global charge vector must vanish across all cells simultaneously. This introduces off-diagonal terms in the action that depend on correlations between cells. The effective action for configuration g_* in the simultaneous sector is:

$$S_{\text{eff}}(g_*) = \frac{Ng_*}{2} - \ln Z_{\text{corr}}(g_*),$$

where Z_{corr} encodes the inter-cell charge correlations. A configuration with richer internal symmetry (higher g_*, more gauge charges to cancel) could have a larger Z_{corr} that compensates the larger S.

Finding

The single-cell measure prefers $N_{\text{gen}} = 1$. But the global constrained measure may prefer a different value if $Z_{\text{corr}}(g_*)$ grows faster than $e^{-Ng_*/2}$. Whether it selects $N_{\text{gen}} = 3$ is a concrete calculation.

What closes OP5

Compute $Z_{\text{corr}}(g_*)$ for the simultaneous emergence configuration as a function of g_* in the constrained path integral (2). If $Z_{\text{corr}}(N_{\text{gen}} = 3) > Z_{\text{corr}}(N_{\text{gen}} = 1)$ by the appropriate exponential factor, TUO derives $g_* = 106.75$. The observable $N_\nu = 2.984 \pm 0.008$ (LEP) is the external check.

7 OP6 — SM Gauge Group from Axioms

Open Problem 6

The SM gauge group SU(3) × SU(2) × U(1) is input to TUO, not derived.

7.1 What TUO constrains

The zero-sum axiom requires that for every gauge generator T^a of the gauge group G:

$$\mathrm{Tr}[\hat{\rho}\,\hat{Q}_{T^a}] = 0.$$

This is the anomaly-cancellation condition. Gauge groups consistent with TUO's zero-sum filter must have representations that allow a matter-only zero-sum configuration.

7.2 Gauge groups satisfying TUO constraints

Group	Rank	Contains SM?	Status
SU(3) × SU(2) × U(1)	4	—	Observed
SU(5)	4	Yes	Predicts proton decay $\tau_p \sim 10^{34}$ yr
SO(10)	5	Yes	$B-L$ is a gauge symmetry; natural
E_6	6	Yes	Exotic; extra colour triplets
$SU(4) \times SU(2)_L \times SU(2)_R$	5	Yes	Pati-Salam

All rows satisfy the anomaly-cancellation conditions and admit matter-only zero-sum configurations. TUO's axioms are *necessary* but not *sufficient* to select the SM gauge group.

7.3 Additional constraints that help

1. **Proton stability.** Super-Kamiokande gives $\tau(p \to e^+\pi^0) > 1.6 \times 10^{34}$ yr. SU(5) at the minimal level predicts $\tau_p \sim 10^{31}$ yr — already excluded. SO(10) with intermediate scales can give $\tau_p > 10^{34}$ yr. This rules out minimal SU(5) from phenomenology, leaving G_{SM}, SO(10), Pati-Salam, and E_6.

2. **Minimality principle.** Among rank-4 groups admitting matter-only zero-sum configurations, G_{SM} is the unique semi-simple group of rank 4 with three factors and one abelian factor. If TUO's constrained PI weights configurations by their gauge complexity (rank, number of generators), then rank-4 groups are preferred

over rank-5, and G_{SM} over SU(5) (different representation content despite same rank).

3. **Yukawa coupling structure.** The SM has the unique property that the top quark mass $m_t \approx 173\,\mathrm{GeV}$ satisfies the fixed-point condition $y_t \approx 1$ at the GUT scale. This is a self-consistency condition that could single out G_{SM} representations.

What closes OP6

Formulate a measure on the space of gauge groups $\{G\}$ satisfying the TUO anomaly-cancellation conditions. The natural candidate is weighting by e^{-S_G} where S_G includes the gauge-kinetic action for the gauge sector in the constrained PI. Combined with the proton-decay experimental constraint, this may select G_{SM} or its minimal GUT extension.

8 Summary and Priority

OP	Problem	Key calculation needed	Difficulty
4	Junction time t_{junc}	2-loop QCD transport, Boltzmann eq. with cooling	**Low**
2	CMB power spectrum	Two-point $\langle T_{\mu\nu} T_{\rho\sigma} \rangle_{\mathrm{ZS}}$, Boltzmann transfer	**Medium**
1	Baryon asymmetry η	Leptogenesis Boltzmann eqs. at T_{TUO}	**Medium**
5	SM content g_*	Inter-cell correlation function $Z_{\mathrm{corr}}(g_*)$	**Hard**
6	SM gauge group	Measure on $\{G\}$ in constrained PI	**Hard**
3	Dark energy Λ	Next-to-leading saddle, topological term	**Very hard**

8.1 New result from this analysis

OP4 Correction

The thermalisation time including FRW cooling is $\tau_{\mathrm{th}} = 152.2\, t_{\mathrm{Pl}}$, revising the junction range from $[17, 94]\, t_{\mathrm{Pl}}$ to $[\mathbf{17.3}, \mathbf{152.2}]\, \mathbf{t}_{\mathrm{Pl}}$. This should be updated in the main paper.

8.2 The hierarchy of difficulty

OP4 can be improved with a single numerical integration using known 2-loop QCD transport coefficients — this is a calculation, not a conceptual advance.

OP2 and OP1 require new calculations in known frameworks (GR perturbation theory and leptogenesis Boltzmann equations) applied to the TUO initial condition $T_{\mathrm{TUO}} = 1.1103\, T_{\mathrm{Pl}}$. The inputs from TUO are clean and precise enough to make these tractable.

OP5 and OP6 require extending the constrained path-integral framework to sum over field content and gauge group choices. This is conceptually new work.

OP3 (dark energy) is the hardest problem in all of physics, reframed in TUO language. The zero-sum axiom provides a structural reason for the tree-level $\Lambda = 0$, which is more than standard QFT offers. But closing the remaining 10^{-123} gap requires a mechanism that no existing framework provides.

8.3 An encouraging observation

Of the six open problems, only OP3 requires genuinely new physics beyond the SM and GR. OP1–OP2 require known physics applied to TUO's initial conditions. OP4 is already partially closed by this analysis ($152\, t_{\mathrm{Pl}}$). OP5–OP6 require extending the constrained PI framework, but the target observable (the LEP measurement $N_\nu = 2.984$, the proton lifetime bound) is already known. This is a better situation than most proposed theories of initial conditions, where the open problems are entirely unconstrained.

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